



NPTEL Online Certification Course

Indian Institute of Technology Roorkee

Course Name: Power System Protection and Switchgear

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WEEK 4: ASSIGNMENT 4

Question 1:

Additional time delay or coordination is generally not required in

- a) carrier blocking scheme.
- b) carrier tripping scheme.
- c) both (a) and (b)
- d) none of these

Answer: (b)

Question 2:

The main disadvantage of a direct underreach transfer tripping scheme is

- a) higher cost.
- b) longer operating time.
- c) maloperation due to noise or switching surges.
- d) requirement of additional time grading.

Answer: (c)

Question 3:

The frequency range of carrier signals transmitted through a microwave communication channel is typically

- a) 0.3-3 Hz
- b) 0.3-3 kHz
- c) 0.3-3 MHz
- d) 0.3-3 GHz

Answer: (d)

Question 4:

Carrier protection schemes are mainly used for

- a) High-voltage transmission lines.
- b) Low-voltage transmission lines.
- c) Distribution lines.
- d) Transformers

Answer: (a)

Question 5:

Pilot-wire relaying schemes are generally used for

- a) short transmission lines.
- b) medium transmission lines.
- c) long transmission lines.
- d) All of these

Answer: (a)

Question 6:

A carrier signal used in protection schemes can also be utilized for

- a) telemetering.

- b) supervisory control.
- c) communication.
- d) All of these

Answer: (d)

Question 7:

A 132 kV long transmission line has an impedance of $(2 + j8) \Omega$. The line is protected by MHO type of distance relay with a characteristics angle of 74° and its first zone covers 80% of the line length. The CT ratio and PT ratio are 500/1 A and 132 kV/110 V, respectively. Considering a fault resistance of 2.4Ω , determine the first zone setting of the distance relay.

- a) $K_1 = 1.27 \Omega$
- b) $K_1 = 2.73 \Omega$
- c) $K_1 = 3.27 \Omega$
- d) $K_1 = 4.73 \Omega$

Answer: (c)

Solution:

Line impedance = $2 + j8 = 8.25 \angle 75.96^\circ \Omega$

CT ratio = 500/1

PT Ratio = 132 kV/110V

Characteristic angle $(\phi) = 74^\circ$

$$Z_1 = (80\% \text{ of line impedance} + \text{fault resistance}) \times \frac{CT \text{ ratio}}{PT \text{ ratio}}$$

$$= (0.8 \times 8.25 \angle 75.96^\circ + 2.4) \times \frac{\left(\frac{500}{1}\right)}{\left(\frac{132 \times 10^3}{110}\right)} = 3.145 \angle 58^\circ \Omega$$

Hence, the first zone setting of mho relay is

$$K_1 = \frac{|Z_1|}{\cos(\theta - \phi)} = \frac{3.145}{\cos(74 - 58)} = 3.27 \Omega$$

Question 8:

A synchronism-check relay is used to

- a) ensure synchronization before closing a circuit breaker.
- b) reduce the impact on the circuit breaker.
- c) maintain system stability.
- d) None of these

Answer: (a)

Question 9:

The use of auto-reclosing improves

- a) stability of the system.
- b) continuity of service.
- c) sensitivity of protection.
- d) None of these

Answer: (b)

Question 10:

Pilot protection provides which of the following advantages?

- a) Simultaneous tripping at both ends of the line
- b) Lower cost compared to conventional protection schemes

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| c) | Both (a) and (b) |
| d) | None of these |

Answer: (a)

****END****